

Program: Diploma in Automation and Robotics	
Course Code:	Course Title: Python Programming Lab
Semester: 3	Credits: 1.5
Course Category: Engineering Science	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To write, test, and debug simple Python programs.
- To implement Python programs with conditionals and loops.
- Use functions for structuring Python programs.
- Represent compound data using Python lists, tuples, and dictionaries.
- Read and write data from/to files in Python.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic concepts of computer programming.		Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	To practice the problem solving approaches in python.	8	Applying
CO2	To use functions for structuring Python programs.	6	Applying
CO3	To use Python data structures – strings, lists, tuples, dictionaries.	8	Applying
CO4	To perform Read and write data from/to files in Python	6	Applying
CO5	To utilize Python packages in developing software applications.	14	Applying
	Lab Exam	3	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			
CO4	3			3			
CO5	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To practice the problem solving approaches in python.		
M1.01	Python IDE installation, adding various packages.	2	Applying
M1.02	Create a program that asks the user to enter their name and their age. Print out a message addressed to them that tells them the year that they will turn 100 years old.	1	Applying
M1.03	Write a python program to find largest number among three numbers	1	Applying
M1.04	Program to convert the given temperature from Fahrenheit to Celsius and vice versa depending upon user's choice.	1	Applying
M1.05	Enter the number from the user and depending on whether the number is even or odd, print out an appropriate message to the user.	1	Applying
M1.06	Program to calculate total marks, percentage and grade of a student. Marks obtained in each of the five subjects are to be input by user. Assign grades according to the following criteria: Grade A: Percentage ≥ 80 Grade B: Percentage ≥ 70 and < 80 Grade C: Percentage ≥ 60 and < 70 Grade D: Percentage ≥ 40 and < 60 Grade E: Percentage < 40	1	Applying

M1.07	Program, using user-defined function to find the area of rectangle, square, circle and triangle by accepting suitable input parameters from user.	1	Applying
CO2	To use functions for structuring Python programs.		
M2.01	Write a python program to find factorial of a given number using functions	1.5	Applying
M2.02	Write a program to create a menu with the following options 1. TO PERFORM ADDITION 2. TO PERFORM SUBTRACTION 3. TO PERFORM MULTIPLICATION 4. TO PERFORM DIVISION Accepts users input and perform the operation accordingly. Use functions with arguments.	1	Applying
M2.03	Write a function that takes a character (i.e. a string of length 1) and returns True if it is a vowel, False otherwise.	1	Applying
M2.04	Implement a Python Program to perform Linear search.	1.5	Applying
M2.05	Implement a Python Program to perform binary search	1	Applying
	Lab Exam I	1.5	
CO3	To use Python data structures – strings, lists, tuples, dictionaries.		
M3.01	Create a list and perform the following methods 1) insert() 2) remove() 3) append() 4) len() 5) pop() 6) clear()	1	Applying
M3.02	Create a dictionary and apply the following methods 1) Print the dictionary items 2) access items 3) use get() 4) change values 5) use len()	1	Applying
M3.03	Create a tuple and perform the following methods 1) Add items 2) len() 3) check for item in tuple 4) Access items	1	Applying

M3.04	Write a python program to check whether the given string is palindrome or not.	1	Applying
M3.05	Take a list, say for example this one: a = [1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89] and write a program that prints out all the elements of the list that are less than 5.	1	Applying
M3.06	Write a python program that accepts a string and calculate the number of upper case letters and lower case letters.	1	Applying
M3.07	Write a program to find sum of all items in a dictionary.	1	Applying
M3.08	Write a python program for filter() to filter only even numbers from a given list.	1	Applying
CO4	To perform Read and write data from/to files in Python		
M4.01	Write a python program to open and write “hello world” into a file.	3	Applying
M4.02	Write a python program to write the content “hi python programming” for the existing file.	1.5	Applying
M4.03	Write a python program to compute the number of characters, words and lines in a file.	1.5	Applying
CO5	To utilize Python packages in developing software applications.		
M5.01	Write a python program to print date, time using date and time functions	1	Applying
M5.02	Write a python program to display a particular month of a year using calendar module	1	Applying
M5.03	Write a python code to read a csv file using pandas module and print the first and last five lines of a file.	2.5	Applying
M5.04	Using a NumPy module create an array of 5 elements.	1.5	Applying
M5.05	Write a NumPy program to create an array of 10 zeros, 10 ones, 10 fives.		
M5.06	To implement a loan calculator using Tkinter	4	Applying
M5.07	Developing a game activity using Pygame	4	Applying
M5.07	LabExam II	1.5	

Online Resources:

Sl No	Website Link
1	https://python-iitk.vlabs.ac.in/
2	https://www.tutorialspoint.com/python/index.htm
3	https://www.w3schools.com/python/
4	https://github.com/

General Instructions:

- Students have to write the date, aim, Software & Hardware requirements for that experiment in the observation book.
- Students have to listen and understand the experiment explained by the faculty and note down the important points in the observation book.
- Students need to write procedure/algorithm in the observation book.
- Analyze and Develop/implement the logic of the program by the student in respective platform
- After approval of logic of the experiment by the faculty then the experiment has to be executed on the system.
- After successful execution the results are to be shown to the faculty and noted the same in the observation book.
- Students need to attend the Viva-Voce on that experiment and write the same in the observation book.
- Update the completed experiment in the record and submit to the concerned faculty in-charge.